B. Subtract Operation

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: data structures, greedy, math, two pointers | https://codeforces.com/problemset/problem/1656/B

You are given a list of n integers. You can perform the following operation: you choose an element x from the list, erase x from the list, and subtract the value of x from all the remaining elements. Thus, in one operation, the length of the list is decreased by exactly 1.

Given an integer k (k gt 0), find if there is some sequence of n-1 operations such that, after applying the operations, the only remaining element of the list is equal to k.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. Description of the test cases follows.

The first line of each test case contains two integers n and k ($2 \le n \le 2 \cdot 10^5$, $1 \le k \le 10^9$), the number of integers in the list, and the target value, respectively.

The second line of each test case contains the n integers of the list $a_1, a_2, \ldots, a_n$ ($-10^9 \le a_i \le 10^9$).

It is guaranteed that the sum of n over all test cases is not greater that $2 \cdot 10^5$.

Output

For each test case, print YES if you can achieve k with a sequence of n-1 operations. Otherwise, print NO.

You may print each letter in any case (for example, "YES", "Yes", "yes", "yEs" will all be recognized as a positive answer).

Example

Input

```
4
4 5
4 2 2 7
5 4
1 9 1 3 4
2 17
17 0
2 17
18 18
```

Output

```
YES
NO
YES
NO
```

Note

In the first example we have the list {4, 2, 2, 7}, and we have the target $k = 5$. One way to achieve it is the following: first we choose the third element, obtaining the list {2, 0, 5}. Next we choose the first element, obtaining the list {-2, 3}. Finally, we choose the first element, obtaining the list {5}.